Máster Universitario en Física y Matemáticas

Trabajo de Fin de Máster



Análisis Bayesiano de Datos Cosmológicos Observacionales

Bayesian Analysis of Cosmological Observational Data

Autor: Alejandro Romero Mora

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Abstract

The ΛCDM model remains the standard framework for describing the evolution of the universe, yet it faces growing statistical tensions between early- and late-universe observables, most notably the Hubble tension. This work investigates the statistical viability of dynamical dark energy (DDE) through the Chevallier-Polarski-Linder (CPL) parametrization, which extends the dark energy equation of state $w(z) = w_0 + w_a z/(1+z)$ beyond the cosmological constant limit ($w_0 = -1$, $w_a = 0$). Using the Bayesian inference framework Cobaya, Markov Chain Monte Carlo (MCMC) sampling is performed combining Cosmic Microwave Background (CMB) data from Planck, Baryon Acoustic Oscillation (BAO) measurements from SDSS and DESI, and Type Ia Supernovae (SNe Ia) datasets, including Pantheon+ and DES Y5 SN. The reconstructed evolution of $w(z)$, the deceleration parameter $q(z)$, and several cosmological distances are analyzed across datasets and redshift ranges to assess how each observable constrains the CPL parameters. The results reveal a mild preference for a time-varying equation of state in the phantom region ($w(z) < -1$) at intermediate and high redshifts, reaching up to 2σ-3σ significance depending on the dataset combination, particularly when DESI BAO measurements are included. The comparison between CMB+BAO and CMB+SNe combinations shows that each probe constrains the (w_0, w_a) plane along different degeneracy directions, determined by their respective pivot redshifts, and that their combination substantially reduces the allowed parameter space. While the results do not conclusively exclude ΛCDM, they identify a consistent pattern that warrants further investigation with upcoming DESI, Euclid, and Vera C. Rubin Observatory data releases, which will help determine whether this preference reflects genuine new physics or a statistical fluctuation in current samples.

Keywords: Dark Energy, CPL Parametrization, Cosmological Parameters, Bayesian Analysis, Cobaya.

Resumen

El modelo ΛCDM constituye el marco estándar para describir la evolución del universo, siendo capaz de explicar de forma consistente tres observaciones independientes - el Fondo Cósmico de Microondas (CMB), las Oscilaciones Acústicas de Bariones (BAO) y las Supernovas de Tipo Ia (SNe Ia) - con un único conjunto de seis parámetros. Sin embargo, este modelo presenta tensiones estadísticas crecientes entre observables del universo temprano y tardío, siendo la tensión de Hubble la más relevante: una discrepancia entre las medidas de H_0 obtenidas a partir del CMB y las SNe Ia (67–68 km/s/Mpc) frente a las medidas locales del universo actual (73–76 km/s/Mpc). Estas discrepancias motivan la búsqueda de física más allá del paradigma estándar, particularmente en la naturaleza de la energía oscura, cuyo origen y dinámica permanecen desconocidos.

Este trabajo tiene como objetivo evaluar la viabilidad estadística de la energía oscura dinámica (DDE) mediante la parametrización Chevallier-Polarski-Linder (CPL), que extiende la ecuación de estado de la energía oscura más allá del límite de constante cosmológica ($w_0 = -1$, $w_a = 0$), definida como $w(z) = w_0 + w_a z/(1+z)$. Esta parametrización, aunque limitada desde el punto de vista teórico al no ofrecer información sobre el origen físico de la energía oscura, resulta muy útil experimentalmente para detectar posibles desviaciones respecto a ΛCDM.

La metodología se apoya en el marco de inferencia bayesiana Cobaya, que mediante algoritmos de muestreo de cadenas de Markov Monte Carlo (MCMC) con la técnica de fast-slow dragging permite ajustar el modelo CPL a distintos conjuntos de datos observacionales: el CMB de Planck, medidas BAO de SDSS y DESI, y catálogos de SNe Ia como Pantheon+ y DES Y5 SN. A lo largo del trabajo se analiza la evolución reconstruida de la ecuación de estado $w(z)$, el parámetro de deceleración $q(z)$, y diversas medidas de distancia —el parámetro de Hubble $H(z)$, la distancia de diámetro angular $d_A(z)$ y la distancia de luminosidad $d_L(z)$— evaluando cómo cada observable y sus combinaciones constriñen el espacio de parámetros del modelo.

Los resultados muestran una preferencia moderada por una ecuación de estado variable en el tiempo, con una tendencia hacia la región fantasma ($w(z) < -1$) a redshifts intermedios y altos, alcanzando una significancia de hasta 2σ–3σ dependiendo de la combinación de datos utilizada, especialmente al incorporar las medidas BAO de DESI2024. La comparación entre las combinaciones CMB+BAO y CMB+SNe revela que cada sonda constriñe el plano (w_0, w_a) en direcciones de degeneración distintas, determinadas por sus respectivos redshifts pivote —las SNe Ia dominan a bajo redshift, mientras que BAO extiende su poder constrictivo hasta redshifts más altos—, de forma que su combinación reduce sustancialmente el espacio de parámetros permitido respecto al uso de un único dataset.

Aunque los resultados obtenidos no permiten excluir de forma concluyente el modelo ΛCDM, sí identifican un patrón consistente que apunta hacia una posible evolución dinámica de la energía oscura, especialmente cuando se incluyen los datos más recientes de DESI. Este hallazgo refuerza la necesidad de seguir investigando con las próximas publicaciones de datos de DESI DR2, Euclid y el Observatorio Vera C. Rubin, que eventualmente proporcionarán la potencia estadística necesaria para

determinar si esta preferencia corresponde a una característica cosmológica genuina o a una fluctuación estadística dentro de las muestras actuales.

Palabras clave: Energía Oscura, Parametrización CPL, Parámetros Cosmológicos, Análisis Bayesiano, Cobaya.

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1 Objectives and methodology

Modern precision cosmology, driven by the synergy between large observational catalogs and the development of numerical inference algorithms, has made it possible to subject cosmological models to unprecedented scrutiny. The concordance model ΛCDM — Lambda Cold Dark Matter —, while providing a robust description of cosmic evolution with only 6 parameters, faces growing statistical tensions between early- and late-universe observables (highlighting the discrepancies in the Hubble parameter H_0). These deviations might suggest the possible existence of physics beyond the standard paradigm, particularly in the nature of dark energy, whose origin and dynamics remain unknown.

The main objective of this work is to assess the statistical significance of dynamical dark energy through the Chevallier-Polarski-Linder (CPL) parametrization, which assumes a smooth variation of the dark energy equation of state with time, and offers great advantages for the study of this component of the universe. The different sources measured in the universe impose constraints that generate discrepancies incompatible with a model as rigid as ΛCDM. This alternative model, thanks to its phenomenological flexibility to test the consistency of ΛCDM, allows for better agreement with observational data.

To carry out this analysis, the methodology relies on the use of the Bayesian inference framework Cobaya (Code for Bayesian Analysis), a Bayesian sampling and cosmological inference code that, through the implementation of Markov Chain Monte Carlo (MCMC) methods, allows fitting the desired model to the datasets of the Cosmic Microwave Background (CMB), Type Ia Supernovae (SNe Ia), and Baryon Acoustic Oscillations (BAO). Mathematical and statistical rigor is verified at each step, and the set of algorithms allows for rapid convergence given the high dimensionality of these types of problems. Through the use of this tool, it will be analyzed how each dataset and their respective combinations impose certain constraints on the model parameters, allowing the quantification of the agreements or discrepancies of CPL with respect to the null model ΛCDM.

2 Brief introduction to cosmology

Cosmology is the branch of physics that studies the origin, the evolution and the structure of the universe. Currently, the most accepted model to describe the universe is ΛCDM (Λ Cold Dark Matter) [Weinberg, 2008]. This represents the simplest mathematical framework that provides a consistent description of the universe's evolution, explaining three independent observations with the same set of parameters.

Under the Cosmological Principle, the geometry of spacetime is described by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric:

$$ds^2 = -c^2dt^2 + a^2(t)\left[\frac{dr^2}{1-Kr^2} + r^2(d\theta^2 + sin^2\theta d\phi^2)\right], \tag{1}$$

where $a(t)$ is the scale factor, representing the expansion of the universe over time. Its current value is fixed to $a_0 \equiv a(t_{present}) = 1$. K represents the curvature, whose

possible values are $K = 0$ for a flat universe, $K > 0$ for a hyperspheric universe and $K < 0$ for a hyperbolic universe. The comoving radial distance is r, and s is the space-time interval, representing the real four-dimensional distance between two objects.

From the expansion of the universe, the light that comes from other galaxies suffers a Doppler effect called cosmological redshift. If a light source emits in a wavelength λ_{em} and is observed on the Earth in λ_{obs}, the redshift z that light undergoes is

$$z = \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} \Longrightarrow 1 + z = \frac{\lambda_{obs}}{\lambda_{em}} \equiv \frac{a_0}{a(t)}. \tag{2}$$

As the redshift is a function of the scale factor, it also depends on time. It means the variables that depend on time can also be expressed as functions of this parameter.

The critical density is defined as the density the universe would have if it were flat:

$$\rho_c(t) = \frac{3H^2(t)}{8\pi G}. \tag{3}$$

One can also define the density parameters as the ratios between the real density of each component and the critical density

$$\Omega_i(z) \equiv \frac{\rho_i(z)}{\rho_c(z)}. \tag{4}$$

The label i represents the different components of the universe (b for baryonic matter, c for cold dark matter, m for total matter, r for radiation and neutrinos and Λ for dark energy). If the universe has curvature, one can also define the curvature density parameter, also from Friedmann equations

$$H_0^2 + \frac{K}{a_0^2} = \frac{8\pi G}{3}[\Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0}]\rho_{c,0} = H_0^2[\Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0}], \tag{5}$$

where the subscript 0 denotes the present-day value of the parameter. Defining the curvature density parameter as

$$\Omega_K \equiv -\frac{K}{a_0^2 H_0^2}, \tag{6}$$

the equation turns into

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_K = 1. \tag{7}$$

Each one of the parameters evolves differently over time. In this context, the equation of state has to be introduced in order to obtain the density evolutions over time. First, w is the parameter of the equation of state, defined as

$$w_i \equiv \frac{p_i}{\rho_i}, \tag{8}$$

where p_i and ρ_i are the pressure and energy density of the ith component. From the conservation equation, one can obtain the relation between density and the scale

factor $a(t)$

$$\dot{\rho} + \frac{3}{a}(p + \rho)\dot{a} = 0 \Longrightarrow \rho(a) = \rho_0 a^{-3(1+w)}. \tag{9}$$

Finally, in terms of the cosmological redshift and the parameter w, the proportionalities are

$$\rho_i \propto (1 + z)^{3(1+w_i)}. \tag{10}$$

In ΛCDM, its values can be derived from general relativity and statistical mechanics:

- Non-relativistic matter: because matter is very diluted, its particles hardly ever collide, so its pressure is insignificant, then $w_m = 0$, and its density dilutes as $\rho_m \propto a^{-3}$.
- Radiation and neutrinos: for a gas formed by massless particles in thermal equilibrium, it can be shown that the average pressure is exactly one-third of the energy density, so $w_r = 1/3$. Radiation dilutes as $\rho_r \propto a^{-4}$.
- Dark energy: since it is associated with vacuum energy, its density has to remain constant (at least by definition), so the only way to achieve it is imposing $w_\Lambda = -1$. In models where dark energy is a dynamic field, called Dynamical Dark Energy (DDE) models, it takes different values. For $-1 < w < -1/3$ (quintessence models), dark energy is interpreted as a scalar field. For $w < -1$ (phantom region), dark energy violates Null (NEC) and Weak Energy Condition (WEC), leading to scenarios that will be explained and studied further on in this work.
- Curvature evolves as $\rho_K \propto a^{-2}$, so it can be seen from (9) that $w = -1/3$.

The interactions of matter and radiation with the geometry of the spacetime are encoded in the Einstein field Equation:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \tag{11}$$

where $G_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu}$ is the Ricci curvature tensor, $R \equiv g^{\mu\nu}R_{\mu\nu}$ is the Ricci scalar, and $g_{\mu\nu}$ is the metric tensor with signature $(-,+,+,+)$. The total energy-momentum tensor $T_{\mu\nu}$ accounts for all matter and energy components in the universe, modeled as perfect fluids:

$$T_{\mu\nu} = (\rho + P/c^2)u_\mu u_\nu + Pg_{\mu\nu}, \tag{12}$$

with ρ and P denoting the total background energy density and isotropic pressure, respectively, and u_μ being the fluid four-velocity.

The Friedmann equations describe the metric expansion of space in homogeneous and isotropic models of the universe.

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2} + \frac{\Lambda}{3}, \tag{13}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P) + \frac{\Lambda}{3}, \tag{14}$$

where G is the Gravitation Constant and Λ is the cosmological constant, that takes account of the acceleration of the expansion of the universe. One of the most representative parameters when studying the history of the expansion is the Hubble parameter $H(z)$. It describes the evolution of the expansion rate, and it can be obtained from these equations. Its present value is the Hubble constant H_0.

Based on these mathematical principles, the most accepted model to describe the origin, evolution and physical properties of the universe is ΛCDM, because it only requires 6 primary parameters to describe the evolution of the universe since the formation of the CMB to the present day:

- **Baryonic matter density** ($\Omega_b h^2$) represents the amount of ordinary matter (protons, neutrons) that forms the gas clouds, stars, planets and other structures. It has influence in the peaks of the BAO.

- **Cold dark matter density** ($\Omega_c h^2$) represents the non baryonic matter, that only interacts gravitationally. It is the responsible for the potential wells in which baryonic matter falls into, enabling the formation of galaxies.

- **Angular size of the sound horizon** (θ_{MC}) is the ratio between the sound horizon (the maximum distance a sound wave could travel in the CMB, denoted as r_s) and the angular diameter distance to the surface of last scattering of the CMB, denoted as d_{sls}. It is a technical measurement of the apparent angular size of fluctuations in CMB, determining the positions of the peaks of the CMB power spectrum.

- **Reionization optical depth** (τ) measures the fraction of CMB photons that were scattered by free electrons created during reionization. It suppresses the amplitude of the CMB at all but the largest scales, and it is the hardest parameter to constrain because its effect is very similar to changing the overall amplitude (A_s), creating a degeneracy that Cobaya must navigate.

- **Overall amplitude** (A_s) describes the amplitude of primordial scalar fluctuations, which enable the formation of galaxies. Increasing A_s raises the entire power spectrum up.

- **Scalar spectral index** (n_s) describes how the anisotropies vary with the scale. If $n_s = 1$, fluctuations are the same at all scales. It varies the height of the higher peaks in relation to the first peak of the CMB power spectrum.

Together, these six parameters allow the model to match the observations from CMB, BAO and supernovae.

The name ΛCDM represents the two dominant components of the current universe's energy density: Λ represents dark energy, a fluid with constant energy density and negative pressure. It is responsible for late-time accelerated expansion of the universe. Its principal drawback is that its physical origin remains unknown.

The Cold Dark Matter (CDM) is a type of matter that does not interact with the electromagnetic spectrum. It provides the gravitational attraction that explains

structure formation in the universe. Without CDM, the small fluctuations seen in the CMB would not have had enough time to collapse into the galaxies observed today.

The model assumes the Cosmological Principle: on sufficiently large scales (typically $> 100Mpc$), the universe is homogeneous (looks the same at every point) and isotropic (looks the same in every direction).

By MCMC estimations, derived parameters can be obtained from the sampled ones [Planck Collaboration, 2020b]. These parameters are usually more discussed in model comparisons, as they are better for comparing with external data and can be better interpreted physically. Some of these parameters are:

- Total matter density (Ω_m) represents the fraction of the universe's energy density that behaves like matter. It is calculated directly from baryonic and cold dark matter density as

$$\Omega_m = (\Omega_b h^2 + \Omega_c h^2)/h^2. \tag{15}$$

- Dark energy density (Ω_Λ). In a flat universe ($\Omega_K = 0$), it is simply calculated as $\Omega_\Lambda = 1 - \Omega_m$. However, in a model with curvature, its value constrains where the model can still explain the expansion measured by supernovae.

- Luminosity distance (d_L). This is the distance obtained in terms of the apparent magnitude m of an object and its absolute magnitude M as

$$M = m - 5log_{10}\frac{d_L}{10pc}. \tag{16}$$

  This is relevant when studying objects with a well-known luminosity as type Ia Supernovae. It can also be related to the comoving distance as

$$d_L = (1+z)\chi. \tag{17}$$

- Angular diameter distance (d_A). This is the distance based on the apparent size of the object in the sky, and it is defined from the real physical size of an object (D) and its angular diameter (θ) as

$$d_A = \frac{D}{\theta}. \tag{18}$$

  It is related to the luminosity distance by the following expression

$$d_A = \frac{d_L}{(1+z)^2}. \tag{19}$$

- Deceleration parameter (q). This parameter measures the cosmic acceleration of the expansion of the universe. It is defined as

$$q = -\frac{\ddot{a}a}{\dot{a}^2}. \tag{20}$$

When $q > (<) \ 0$, the universe is decelerating (accelerating). Then, when the parameter is $q = 0$, the universe is changing from acceleration to deceleration (or vice versa)

- The comoving angular diameter distance (D_M) represents the transversal comoving distance between two objects separated by a certain angle in the sky. This is the distance that connects the geometrical measurements with the observable distances d_A and d_L. For a flat universe, it is mathematically defined as the integral of the inverse of the Hubble parameter:

$$D_M(z) = \int_0^z \frac{c}{H(z')} dz'. \tag{21}$$

  This distance acts as the projection factor that maps the comoving sound horizon (r_s) to its observed angular size (θ_*) on the sky. Consequently, it serves as a fundamental geometric bridge to connect early-universe CMB measurements with the late-time expansion history of the cosmos.

ΛCDM, along with the set of parameters, is able to explain three independent observations from the universe:

- The Cosmic Microwave Background (CMB): in the beginning, the universe was a hot and dense plasma formed by protons, electrons and photons, which could not travel long distances because they were colliding with free electrons, so it is said that they remained confined. As the universe expanded, it was cooling down, and when the temperature dropped to around $3000K$, several events took place: protons and electrons combined to form hydrogen atoms (recombination), so photons stopped colliding with the electrons and started travelling freely (decoupling). The surface of photons' last scattering formed the CMB, a near-perfect black body emission, with little temperature oscillations called anisotropies.

  Many of the characteristics of the universe are encoded in the CMB power spectrum represented in Figure 1, which is obtained by the CMB observations and subsequent foreground cleaning and decomposition into spherical harmonics. This process reduces the information of millions of pixels into a set of multipoles (ℓ) that capture the statistical properties of the early universe. The positions, heights and proportions between the different peaks are closely related to the parameters and the physical properties of the universe, so it is essential to analyze how a given set of parameters fits the spectrum.

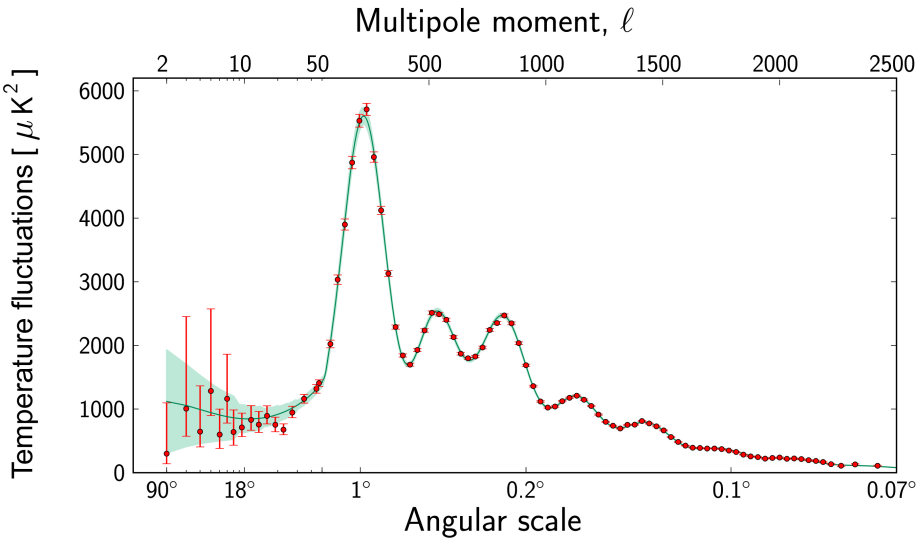


Figure 1: CMB power spectrum. The red dots are measurements made with Planck, shown with error bars that account for measurement errors and an estimate of the uncertainty that is due to the limited number of points in the sky. The green curve represents the best fit of the 'standard model of cosmology', and the pale green area around the curve shows the predictions of all the variations of the standard model that best agree with the data.

- Baryonic Acoustic Oscillations (BAO): In the primordial plasma, there were two opposing forces. Dark matter and baryons tended to collapse gravitationally in the zones with higher density. However, photons were so closely coupled to baryons, generating an outward radiation pressure that resisted the collapse. This generated outward-travelling pressure (sound) waves called BAO, and in the drag epoch ($z \sim 1060$), when radiation pressure vanished, these sound waves abruptly froze. The distance traveled by the waves from the initial perturbations to the drag epoch is the so-called sound horizon (r_s). Over time, dark matter fell into the baryonic shells while baryons collapsed into the central dark matter overdensities, leaving a characteristic signature in the cosmic large-scale structure. Due to the scale of the sound horizon, it is statistically more likely to find two galaxies separated by a comoving distance of ~ 150 Mpc than any other configuration.

- Type Ia Supernovae: A supernova is a colossal explosion of a star at the end of its life. This event emits a large amount of energy, and it occurs because the thermal pressure can no longer counteract the gravitational pull generated by its mass, so it collapses creating an energetic explosion. There are different types, but the most useful in this context are the so-called type Ia Supernovae (SNe Ia). This type of SNe occurs in binary stellar systems, having a white dwarf and another star. As the white dwarf is more dense than a star at another stage of its life, the degeneracy pressure of the electrons is very high.

This is translated into a higher surface gravity, so when the companion star evolves into a red giant and expands beyond its Roche lobe (the region around a star where its mass remains anchored to it), matter begins to transfer onto the white dwarf. This type of star has a mass limit that cannot be exceeded because of Pauli's Exclusion Principle: when density is very high, electrons have to occupy more energetic states, so they move at ultra-relativistic speeds. When the mass exceeds this limit, called the Chandrasekhar limit, the electron degeneracy pressure is no longer able to counteract gravity, so it collapses and generates a supernova when its mass is $M_{CH} \approx 1.44 M_{\odot}$. As a result, all SNe Ia reach a similar peak intrinsic luminosity, so measuring their apparent luminosity it is possible to determine their luminosity distance, making them useful as standard candles.

However, this model presents some theoretical and observational problems, which suggest that it is necessary to reconsider or modify it. One of the problems is the cosmological constant (Λ), the responsible for the acceleration of the expansion of the universe, whose origin and nature remains unknown, in addition to inconsistencies like the Hubble Tension [Riess et al., 2022], a discrepancy between the measurements of the Hubble constant H_0 from the CMB and type Ia Supernovae ($67 - 68 km/s/Mpc$) and local measurements from the current universe ($73 - 76 km/s/Mpc$). These tensions are the motivations to discover new physics beyond this model by improving the quality of the measurements and the statistical models.

Adding new parameters to the model (such as the spatial curvature Ω_K or parameterizing the dark energy equation of state w), allows the six primary parameters to adjust in order to keep the model consistent with the data, resulting in alternative theories.

2.1 Cosmological Perturbation Theory

So far, the universe has been treated as perfectly homogeneous. However, to understand the formation and evolution of large-scale structures, inhomogeneities have to be introduced. As long as they remain relatively small, they can be treated using perturbation theory. The procedure will involve expanding the Einstein equations order-by-order in perturbations to the metric and the stress tensor.

Before obtaining the equations, it is necessary to choose a Gauge. CAMB uses the Synchronous gauge following the formalism of Ma and Bertschinger [1995]. This allows for a direct correspondence between the metric perturbation variables (h_{ij}, η) and the numerical calculation of the transfer functions. In this gauge, the metric is written as

$$ds^2 = a(\tau)^2[-d\tau^2 + (\delta_{ij} + h_{ij})dx^i dx^j]. \tag{22}$$

The metric perturbation h_{ij} can be decomposed into a trace term $h \equiv h_{ii}$, and three traceless components, $h^{\parallel}_{ij}$, $h^{\perp}_{ij}$, which divergences are respectively longitudinal and transverse by definition, and h^{T}_{ij} is transverse, satisfying

$$\epsilon_{ijk}\partial_j\partial_l h^{\parallel}_{lk} = 0, \qquad \partial_i\partial_j h^{\perp}_{ij} = 0, \qquad \partial_i h^{T}_{ij} = 0,$$

where $h_{ij} = h\delta_{ij}/3 + h^{\parallel}_{ij} + h^{\perp}_{ij} + h^{T}_{ij}$. Then, $h^{\parallel}_{ij}$ can be written in terms of a scalar field μ and a divergenceless vector $\vec{A}$ as

$$h^{\parallel}_{ij} = \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)\mu,$$
$$h^{\perp}_{ij} = \partial_i A_j + \partial_j A_i, \qquad \partial_i A_i = 0.$$

In the Fourier space, for a given mode with wavenumber k aligned along the z-axis, the metric perturbation can be decomposed in two scalar fields, $h(\vec{k},\tau)$ and $\eta(\vec{k},\tau)$ as follows:

$$h_{ij}(\vec{k},\tau) = \hat{k}_i\hat{k}_j h + \left(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij}\right)6\eta, \qquad \vec{k} = |k|\hat{k}.$$

The Einstein equations relate the geometry of the space with the distribution of matter, represented by the Einstein tensor and the energy-momentum tensor respectively. Through the metric perturbation, this set of equations serve as a bridge between the mathematics of cosmic perturbations and a computationally feasible numerical model.

$$\delta G^{\mu}_{\nu} = 8\pi G\delta T^{\mu}_{\nu}. \tag{23}$$

There are four independent linearized Einstein equations in CPT, but the ones of greatest interest to study in this context are the following (using the $(+,-,-,-)$ convention for the metric):

- Component 0-0:

$$k^2\eta - \frac{1}{2}\mathcal{H}\dot{h} = -4\pi G a^2\delta\rho_{tot}. \tag{24}$$

- Traceless component i-j:

$$\ddot{h} + 6\ddot{\eta} + 2\mathcal{H}(\dot{h} + 6\dot{\eta}) - 2k^2\eta = -24\pi G a. \tag{25}$$

Where $\mathcal{H} \equiv \frac{\dot{a}}{a}$ is the conformal Hubble parameter and dots denote derivatives respect to τ, $\dot{a} \equiv \frac{\partial a}{\partial\tau}$. For convention, $\theta = ik\cdot v$ is the velocity divergence. For each component, these equations have to be solved. Let us define the density fluctuation as $\delta \equiv \frac{\rho-\bar{\rho}}{\bar{\rho}}$. Each one of the components of the universe satisfy its continuity and Euler equations.

Cold Dark Matter: as it is a non-collisional and cold particle ($p_{cdm} = 0, v_{cdm} = 0$), in the synchronous gauge its speed can be fixed to 0 ($\theta_{cdm} = 0$), simplifying its continuity equation to

$$\dot{\delta}_{cdm} = -\frac{1}{2}\dot{h}. \tag{26}$$

Baryons are coupled to photons before decoupling, moving with the speed of sound in the baryonic fluid, c_s:

$$\dot{\delta}_b = -\theta_b - \frac{1}{2}\dot{h}, \tag{27}$$

$$\dot{\theta}_b = -\mathcal{H}\theta_b - k^2c_s^2\delta_b + \frac{4\rho_\gamma}{3\rho_b}an_e\sigma_T(\theta_\gamma - \theta_b). \tag{28}$$

The last term represents the interaction by Thomson dispersion between free electrons and photons in the plasma.

The dark energy continuity equations $\delta_{de}(k,\tau)$ and $\theta_{de}(k,\tau)$ depend on the model in study, so its expressions will be developed for the CPL parameterization case in Section 2.2.

The evolution of relativistic species (photons and neutrinos) in the synchronous gauge is described by the relativistic Boltzmann equation for the k-space distribution function $f(x,p,\tau)$. Expanding the directional dependence $\mu = \hat{\vec{k}} \cdot \hat{n}$ in terms of Legendre polynomials $P_\ell(\mu)$, the continuum distribution is projected onto a set of multipole moments $F_\ell(k,\tau)$, this step can be done because the perturbation function F_ν is continuous in the compact interval $\mu \in [-1,1]$:

$$F_\nu(\vec{k},\hat{n},\tau) = \sum_{\ell=0}^{\infty}(-i)^\ell(2\ell+1)F_{\nu\ell}(\vec{k},\tau)P_\ell(\vec{k}\cdot\hat{n}). \tag{29}$$

Using the orthogonality property of Legendre polynomials, $\int_{-1}^{1} P_\ell(\mu)P_{\ell'}(\mu)d\mu = \frac{2}{2\ell+1}\delta_{\ell\ell'}$, and projecting the continuous function $F_\nu(\vec{k},\hat{n},\tau)$ onto the discrete space of multipolar coefficients $F_\ell(k,\tau)$, one can obtain a set of equations that can be truncated:

$$F_\ell(k,\tau) = \frac{i^\ell}{2}\int_{-1}^{1} F_{\nu\ell}(k,\mu,\tau)P_\ell(\mu)d\mu. \tag{30}$$

The lower-order multipoles encode direct physical quantities: the monopole ($\ell = 0$) represents the density perturbation δ, the dipole ($\ell = 1$) corresponds to the velocity divergence θ, and the quadrupole ($\ell = 2$) measures the anisotropic shear σ. Multipoles with $\ell \geq 3$ account for high-order spatial anisotropies during free-streaming.

Applying the recurrence relations of Legendre polynomials yields the coupled system of differential equations:

$$\dot{F}_\ell = \frac{k}{2\ell+1}[\ell F_{\ell-1} - (\ell+1)F_{\ell+1}] + \mathcal{S}_\ell, \tag{31}$$

where $\mathcal{S}$ contains the metric perturbations and collision terms (such as Thomson scattering for photons). Given that this hierarchy is infinitely dimensional, the spectrum is calculated up to $\ell'_{max} = \ell_{max} + \Delta\ell$, where $\Delta\ell \sim 200 - 500$. Then, the system can be numerically truncated to a maximum number of poles ℓ_{max} without problems around the boundary. This is the procedure followed by CAMB and CLASS, the Boltzmann solvers integrated in Cobaya, that will be explained in greater detail in C.

Taken together, these equations constitute the indispensable bridge between cosmological theory and the observational data of the primordial universe, by transforming the continuous physics in phase space into a finite set of ODEs. Thus, Boltzmann solvers can compute the evolution of each individual multipole, thereby generating the anisotropy power spectra from the CMB (C_ℓ) and large-scale structure ($P(k)$). These predictions can be directly compared with the high-precision observations from CMB, BAO and SNe Ia, enabling the execution of Bayesian in-

ference algorithms and MCMC sampling to constrain the fundamental parameters of the cosmological model.

2.2 CPL parameterization

Some of the most studied alternative models are the ones where the dark energy equation of state (w) is a new parametrized variable instead of constant and equal -1 as in ΛCDM. These are called Dynamical Dark Energy (DDE) models. One of them is the Chevallier–Polarski–Linder (CPL) parameterization [Chevallier and Polarski, 2001, Linder, 2003], where the equation of state is given by only two independent parameters that capture the evolution of the dark energy through the history of the universe. Mathematically, it is defined as

$$w(z) = w_0 + w_a \frac{z}{1+z}. \tag{32}$$

Due to the mathematical form of the parameterization, the equation of state is well delimited in the early universe, tending to a well-determined finite value when $z \to \infty$. Its simplicity - by only adding 2 new parameters - allows the MCMC algorithm to break the degeneracies and to converge faster than with a more complex model. In addition, CPL concentrates all its sensitivity in the range $z \sim 0-2$, where the dark energy plays an important role in the universe, cause it is a transition era, and it is also the range in which BAO and Supernovae are mostly measured.

While CPL provides a robust framework for the expansion history of the universe, it is strictly a background-level parameterization. Consequently, it does not consistently define the perturbative degrees of freedom required to solve the linear perturbation equations. These properties must be specified explicitly in the numerical solvers. Then, even when this model is poor from the perspective of theoretical physics because it does not give information about the origin or the nature of dark energy, experimentally it is great to detect where ΛCDM has weaknesses.

In this model, the Hubble parameter can be obtained from the Friedmann equation - assuming the universe is flat and without radiation - as:

$$H^2(z) = H_0^2[\Omega_m(1+z)^3 + \Omega_\Lambda(z)], \tag{33}$$

where $\Omega_\Lambda(z) \propto (1-\Omega_m)\rho_\Lambda(z)$. To obtain the analytical expression of the dark energy density, the energy conservation equation can be used:

$$\frac{d\rho}{dz} = \frac{3(1+w(z))}{1+z}\rho. \tag{34}$$

Substituting the equation of state and integrating, one can obtain

$$log\rho_\Lambda \propto 3\int_0^z dz' \frac{1+w_0+w_a\frac{z'}{1+z'}}{1+z'} = 3\left[(1+w_0+w_a)log(1+z) + w_a\frac{z}{1+z}\right]. \tag{35}$$

Then,

$$\rho_\Lambda \propto (1+z)^{3(1+w_0+w_a)} e^{-3w_a\frac{z}{1+z}}. \tag{36}$$

Substituting in (33), one can obtain an analytical expression for Hubble parameter

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0+w_a)}e^{-3w_a\frac{z}{1+z}}}. \tag{37}$$

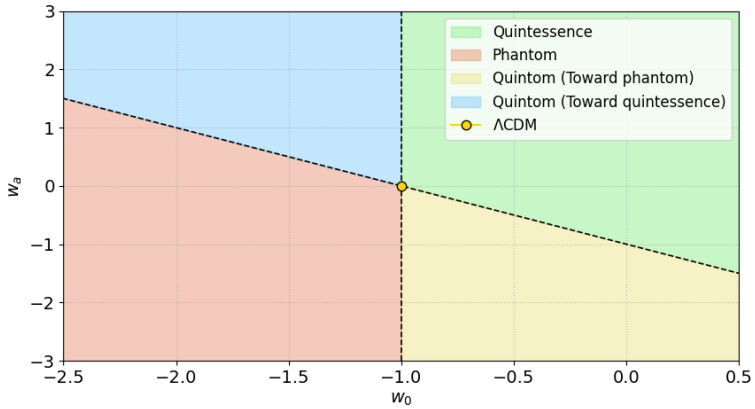


Figure 2: Regimes of the CPL dark energy equation of state.

This model allows different regimes depending on the values of (w_0, w_a) [Copeland et al., 2006]. The theoretical and observational reference is the ΛCDM limit, which occurs when $(w_0, w_a) = (-1, 0)$, corresponding to a cosmological constant Λ. If $w(z) \geq -1$, the model is classified as quintessence. In this region, the equation of state satisfies the Dominant Energy Condition (DEC) throughout cosmic history, $w_0 \geq -1$ and $w_0 + w_a \geq -1$, and the energy density gradually dilutes as the universe expands, making it a theoretically stable regime. Figure 2 shows all the regions of the corresponding regimes in the plane (w_0, w_a).

In the phantom region, where $w(z) < -1$, dark energy violates the DEC, since $w_0 < -1$ and $w_0 + w_a \leq -1$. This implies having an energy density that increases with the expansion of the universe, leading to a ultimate fate known as the Big Rip, where galaxy clusters, solar systems, and, ultimately, subatomic structures are torn apart. Numerically, this causes divergences that have to be regularized as $w(z) \to -1$.

When the equation of state crosses at some point in cosmic history from the Phantom to the quintessence region or vice versa, these are quintom models or models with Phantom Crossing.

The dark energy ODEs for the CPL perturbations are defined using two different sound speeds. The adiabatic sound speed is directly imposed by the temporal derivative of the equation of state, $c_a^2(a) \equiv \frac{\dot{\bar{p}}_{DE}}{\dot{\bar{\rho}}_{DE}}$. The effective sound speed in the rest frame of the fluid is a free parameter that describes the resistance of the fluid to cluster on microphysical scales. It is usually fixed to $c_s^2 = 1$ (the speed of light). The equations that describe the dark energy fluctuations are:

$$\dot{\delta}_{\mathrm{DE}} = -(1+w)\left(\theta_{DE} + \frac{\dot{h}}{2}\right) - 3\mathcal{H}(c_s^2 - w)\delta_{DE} - 9\mathcal{H}^2(1+w)(c_s^2 - c_a^2)\frac{\theta_{DE}}{k^2}, \tag{38}$$

$$\dot{\theta}_{DE} = -\mathcal{H}(1 - 3c_s^2)\theta_{DE} + \frac{c_s^2 k^2}{1+w}\delta_{DE}. \tag{39}$$

This set of equations establishes the background and linear perturbation framework implemented in this work, allowing the CPL parametrization to be constrained against cosmological data within the Cobaya sampling environment.

3 Cosmological data

To achieve the best possible results, in addition to a good model that ensures statistically and physically reliable results, a good and diverse set of data is also needed, to be able to cover all the possible degeneracies using different combinations, and also to compare its predictions separately, looking for inconsistencies. Cobaya uses datasets that collect data from different objects in the universe, and then the likelihood evaluates the difference between the observed data and the point predicted by the chain. Therefore, it is important to be familiarized with them in order to understand and analyze the results they produce.

3.1 CMB datasets

While various ground-based experiments such as the Atacama Cosmology Telescope (ACT) and the South Pole Telescope (SPT) offer enhanced angular resolution at high multipoles ($\ell \gtrsim 3000$), the primary Cosmic Microwave Background (CMB) likelihoods in this work are drawn exclusively from the legacy data release of the Planck satellite [Planck Collaboration, 2020a]. CMB observations from space remain unaffected by atmospheric noise and systematic uncertainties inherent to ground-based observations, delivering full-sky measurements across both temperature and polarization angular power spectra. Because Planck provides the most robust full-sky baseline for the acoustic peak structure at intermediate and large scales, where cosmological background dynamics and late-time dark energy effects primarily leave their signatures, it serves as the definitive reference dataset.

Planck was a space observatory developed by the European Space Agency (ESA) that operated from May 2009 until 2013. Its principal objective was to map the anisotropies of the CMB with an unprecedented angular resolution, sensitivity and frequency coverage.

To isolate the primordial CMB signal from the foreground contamination of the Milky Way, Planck imaged the entire sky simultaneously across 9 different frequency channels from $30GHz$ to $857GHz$.

The spacecraft operated from the Lagrange point L_2, allowing it to keep the Sun, Earth and Moon behind its thermal shields while scanning the deep cosmos to avoid thermal and electromagnetic radiation.

The Planck dataset is a massive public data ecosystem that is still being explored. It is compiled in the Planck Legacy Archive (PLA), maintained by ESA. The collaboration released its data in three major increments: 2013, 2015 and the definitive 2018 (Legacy) release. Since then, advanced re-processings have refined the polarization data.

To have computationally manageable data, it is necessary to transform the initial spherical map into spherical harmonics, helping to reduce millions of pixels into a vector with 2500 values, without losing any relevant physical information.

The Planck power spectra are categorized into three main physical observables:

1. TT (Temperature-Temperature) measures the subtle temperature variations in the primordial plasma just before recombination ($z \approx 1100$). Cobaya extracts the acoustic peaks, determining critical parameters such as the geometry of the universe from the first peak (confirming $\Omega_k \approx 0$), and the exact proportion between baryons from the second and the third peak.

2. E-mode polarization is generated by the Thomson scattering of a quadrupole temperature anisotropy around free electrons. This means the electron has to be surrounded by photons that are hotter in one direction and colder in the perpendicular direction. As temperature and E modes are correlated because they are generated by the same density perturbations, TE modes measure this statistical correlation. EE modes contain the autocorrelation of this polarization map. Analyzing them together, TE and EE modes contain information about the reionization of hydrogen, helping to determine the optical depth as an independent parameter, and consequently unlocking the exact value of A_s.

3. Polarized lensing: As the universe is not homogeneous or isotropic, photons from CMB have been deflected from their straight trajectory due to gravitational lensing. Planck is able to measure the power spectrum of this lensing, obtaining the abundance of massive structures depending on their size. This data contains information from the late universe ($z \approx 1 - 4$), helping to determine parameters related to matter densities, such as w in DDE models.

3.2 BAO datasets

BAO datasets are fundamental to map the late history of the expansion and the dark matter distribution in the universe, giving fundamental information about how the expansion has been over the last billions of years.

To build the BAO datasets, the telescope measures the angular positions and the redshifts of millions of galaxies, being able to create a three-dimensional map where every point represents a galaxy. To extract BAO signals, it is necessary to use the two point correlation function, that measures the probability to find a galaxy from a certain distance from a given one. Due to the sound waves in the plasma, there is an excess of probability to find galaxies separated by exactly $150 Mpc$.

By measuring the statistical separation between millions of galaxies, one can obtain the angular diameter distance ($D_A(z)$) from different points for a given redshift, and the Hubble parameter ($H(z)$) measuring the separation for different redshifts. The evolution of Hubble parameter for different periods allows the model to know exactly how is the expansion of the universe at every moment. As Planck does not directly constrain this parameter, BAO datasets break the degeneracy between the Hubble constant and the dark energy equation of state, and in dynamic models like CPL, they give almost all the power of the statistical analysis, just because the

acceleration of the universe occurs in the range $z \sim 0.3 - 1.5$, where the greatest constraining power is concentrated in these datasets.

The first generation of datasets was released after detecting the first signs of the $150Mpc$ bubble in 2005. Then, 6dFGS (Six-Degree-Field Galaxy Survey) mapped the southern sky, and SDSS swept across the northern sky, providing data at a very low redshift. This is useful to fix the initial conditions, ensuring the model will not diverge in this range.

In 2014-2016, BOSS (Baryon Oscillation Spectroscopic Survey) [Alam et al., 2017] measured spectroscopically more than 1.2 million galaxies. This was the first dataset able to separate the angular diameter distance and the Hubble distance.

Then, the dataset SDSS (also known as eBOSS) [Alam et al., 2021] was released in 2020, adding quasars to the measurements. In the center of most galaxies, there is a supermassive black hole, with the power to devour the matter it finds in its surroundings and captures with its gravitational pull. This creates an accretion disk with a brightness that can be thousands of times higher than the entire galaxy. When this happens, it is said that the black hole is active, so it is called a quasar. Using its brightness, SDSS used quasars from high redshifts ($z \sim 1.1 - 2.2$) as if they were normal galaxies.

For the farthest ones, SDSS used quasars as light sources. In the universe, there are neutral hydrogen clouds (HI), the most abundant element in the universe. The difference between the ground state and the first excited state of its electron is $10.2eV$, the energy of a photon with $1215.67\mathring{A}$. This transition is called $Lyman - \alpha$. Then, if a white light passes through one of these clouds, the exact wavelength associated with the excitation is absorbed, leaving its mark on the spectrum. As the universe is in a continuous expansion, the wavelength of the mark changes, so if the light passes through another cloud, it will leave another mark. When light from a far quasar is detected, the spectrum contains the density (from its degree of attenuation) and size (from its wavelength) of every cloud through which the light has passed, creating the so-called Lyman-α forest. By this way, SDSS was able to measure not only distant quasars, but also these invisible clouds that cannot be detected by the light they emit. This is essential to improve the accuracy of data in the universe for $z \sim 2.1 - 3.5$, where the expansion of the universe was still not accelerating.

In 2024, DESI published a dataset containing its first year of observations, with more than 5.7 million unique spectra, called DESI2024 [Adame et al., 2025]. Its greatest achievement is measuring the BAO scale with a precision of more than 0.5%, helping to constrain parameters of interest like $H(z)$, and consequently reducing the allowed parameter space in models like CPL. When combining DESI2024 with Planck and a SNe Ia dataset, a statistical preference for DDE models with an evolution of dark energy toward the phantom region was suggested, which will be examined in the further analysis of the results. DESI DR2 [Abdul Karim et al., 2025] is the natural evolution of the experiment, accumulating 3 years of measurements and consequently reducing the uncertainties. Its results could serve to confirm or reject the preference of the data for DDE models.

3.3 Supernovae datasets

Ia type Supernovae constitute one of the pillars in the study of the Hubble tension [Riess et al., 2022], as they can be linked with the local distance ladder. In addition, they are responsible for the discovery of the accelerated expansion of the universe [Riess et al., 1998].

As the critical mass (M_{CH}) is almost the same for every SNe Ia, the brightness they emit is also the same, so they can be used as standard candles. The supernova emits energy for the next months after the explosion, with a maximum peak 20 days after it happens, and then it starts decreasing. By only measuring their maximum apparent magnitude (m_B) and the redshift (distance) to the source, one can infer how has been the evolution of the expansion from the moment of the emission to the moment it is detected. Consequently, this allows obtaining the evolution of the luminosity distance, using the following relation

$$m_B - M = 5log_{10}\frac{D_L(z)}{10pc}, \tag{40}$$

where M is the real absolute brightness. In the practice, Cobaya compares the distance modulus $\mu(z) = m_B^{standardized} - M$ with the one obtained by the theoretical luminosity distance integral that comes from Friedmann equations. Actually, not all SNe Ia have exactly the same intrinsic brightness, so the first datasets used the Tripp relation to correct these differences, defined as

$$m_B^{standardized} = m_B + \alpha \cdot x_1 - \beta \cdot c, \tag{41}$$

where x_1 is the stretch factor, that measures how many days the curve takes to rise and fall, c is the color factor that measures how red or blue is the supernova on its maximum brightness, and α and β are correlation coefficients. However, in modern articles and datasets, this equation has been modified introducing a statistical correction term determined by Monte Carlo simulations, needed to account for selection effects and other issues in distance recovery. This non linear function $\delta_{\mu-bias}$ depends on z, x_1 and c, and the modified Tripp relation is defined as

$$m_B^{standardized} = m_B + \alpha \cdot x_1 - \beta \cdot c + \delta_{\mu-bias}(z, x_1, c). \tag{42}$$

The first SNe Ia datasets were small, and every group used different filters and calibrations. The first time the community rigorously standardized the data was in 2010-2014, with the project JLA (Joint Light-curve Analysis), with a sample of 740 Supernovae. JLA demonstrated that the stretch and colour factor have to be treated as nuisance parameters inside the MCMC, so they can be marginalized when analyzing the results.

In 2018, Pantheon unified the data from JLA with more recent datasets such as Pan-STARRS1, HST and SNLS, cataloging a total of 1048 SNe Ia in the range $z \sim 0.01 - 2.26$. The most important achievement of Pantheon was reducing the systematic errors in photometric calibrations.

In 2022, Pantheon+ [Scolnic et al., 2022, Brout et al., 2022], using a total of 1701 SNe Ia, linked them with the local distance ladder using Cepheids, enabling

the study of the Hubble tension.

In 2024, Dark Energy Survey (DES) completed the biggest homogeneous SNe Ia dataset ever released, called DES Y5 SN or DES-Dovekie Y5 SN [DES Collaboration, 2024, Popovic et al., 2026], with more than 1600 supernovae measured with the same instrument and processed with the same software. DES Y5 SN also introduced Machine Learning techniques to identify SNe Ia.

4 Results

This section details the primary findings of our MCMC parameter estimation, focusing on the dynamic response of the expansion history to different data combinations. We systematically examine the reconstructed redshift evolution of the dark energy equation of state $w(z)$ and its direct influence on different parameters of interest. Finally, we investigate how sequentially combining observational likelihoods restricts the available parameter space, detailing the shrinkage and shift of the marginal covariance ellipses in the w_0–w_a plane relative to the standard ΛCDM baseline.

4.1 $w(z)$ and the phantom region

When extending the cosmological framework to accommodate a DDE equation of state via the CPL parameterization, the results obtained show a preference for behaviors beyond the standard model. These results must be analyzed carefully to avoid making definitive claims without having sufficient statistical significance.

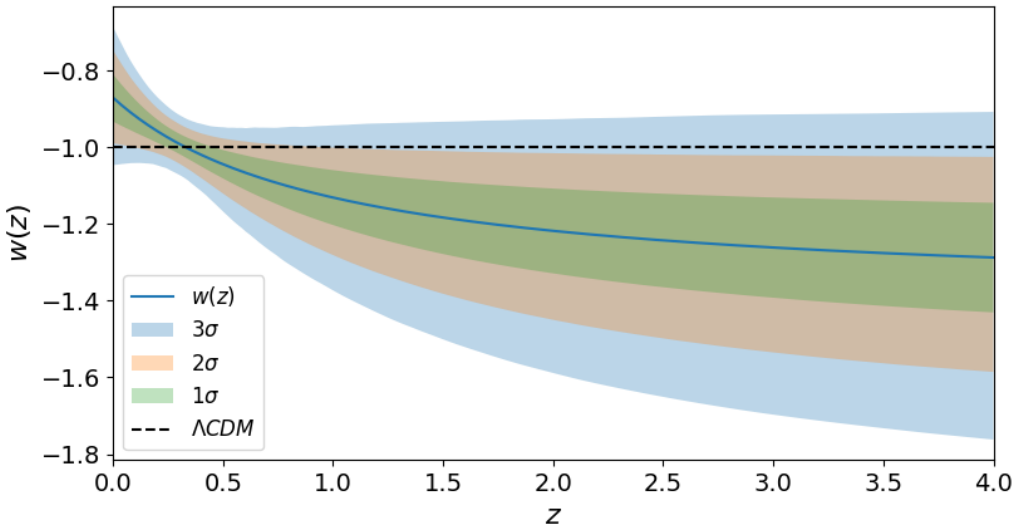


Figure 3: $w(z)$ with 1σ, 2σ and 3σ error bars. The chain contains the combination of Planck, SDSS and Pantheon+ likelihoods

As observed in Figure 3, the equation of state $w(z)$ within the CPL framework shows a preference for the phantom region at high redshifts. For $z \gtrsim 1$, the 2σ confidence interval lies entirely within the phantom region $w(z) < -1$, and only

when expanding the uncertainty up to 3σ, the parameter space allows a non-phantom behavior. Then, the results show a phantom crossing at 2σ significance.

A 2σ exclusion for ΛCDM baseline represents a mild tension, but it must be treated with caution. From a statistical standpoint, 2σ does not represent a robust discovery, as it remains susceptible to statistical fluctuations within the dataset, such as random noise or systematical errors in the datasets. In addition, the CPL parameterization is a "toy model" designed to observe the behavior of dark energy in the late universe, that asymptotically approaches $w_0 + w_a$ at high redshifts, so the evolution of the equation of state is governed by its mathematical structure. Consequently, a preference for non-zero values of w_a at mid redshifts can artificially project a phantom trajectory.

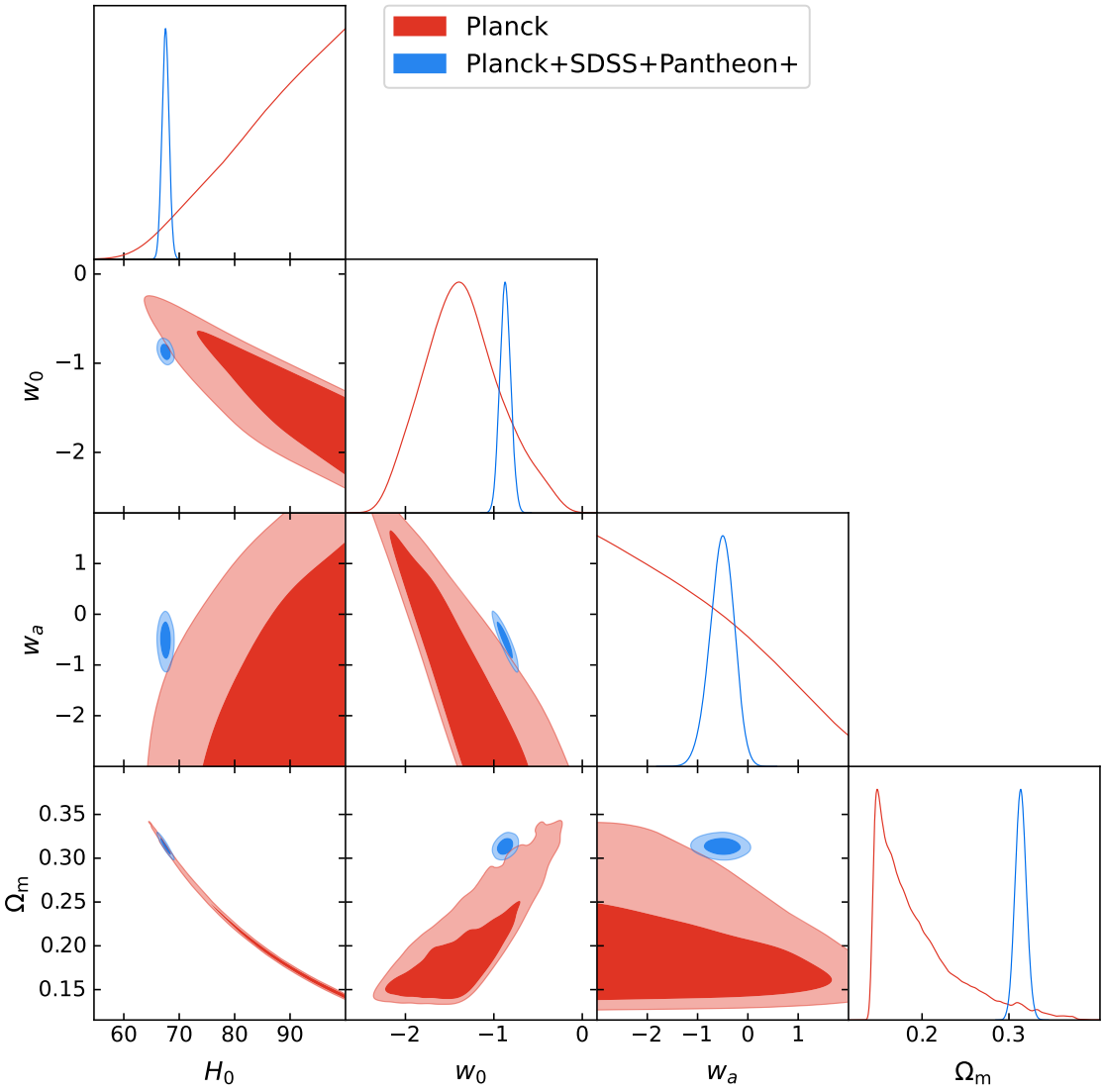


Figure 4: Triangular plot of some of the parameters of interest. The blue ellipses represents the results from the combination of Planck, SDSS and Pantheon+, and the red ones, only from Planck [Lewis, 2019].

It is important to mention that the CMB likelihood alone provides no direct

constraint on the DDE parameters (w_0, w_a), since the epoch of recombination ($z \sim 1100$) only provides the integrated expression of the comoving angular diameter distance, that indirectly depends on both of them through the Hubble parameter. However, this constraint is degenerated, so its primary contribution on the dark energy equation of state comes from fixing the physical matter density (Ω_m) and the scale of the sound horizon (r_s) and, as a result, breaking the degeneracies between the BAO scale and the cosmological parameters. In Figure 4, the degeneracies between w_0, w_a, H_0 and Ω_m for the Planck dataset are clearly visible through the elongated shape of the ellipses, and the insufficient convergence can be noticed in some plots such as the (w_0, w_a) projections, where contours do not close and extend beyond the plotted boundaries.

4.2 CMB+BAO vs. CMB+SNe

Due to their different nature and measurement methods, the SNe Ia and the BAO datasets show slightly different results, which is an advantage when they are combined. In this case, the datasets will be analyzed separately, to assess how each one constrains the evolution of the parameters of interest. To conduct this comparison rigorously, all the chains will contain the CMB likelihood to avoid degeneracies. To parameterize the dynamical behavior of dark energy, the CPL model has been adopted. Four independent MCMC chains were generated with different SNe Ia (DES-Dovekie Y5 SN and Pantheon+) and BAO (SDSS and DESI DR2) datasets. First of all, it is optimal to analyze together the deceleration parameter evolution and the Hubble parameter evolution, as they provide complementary information.

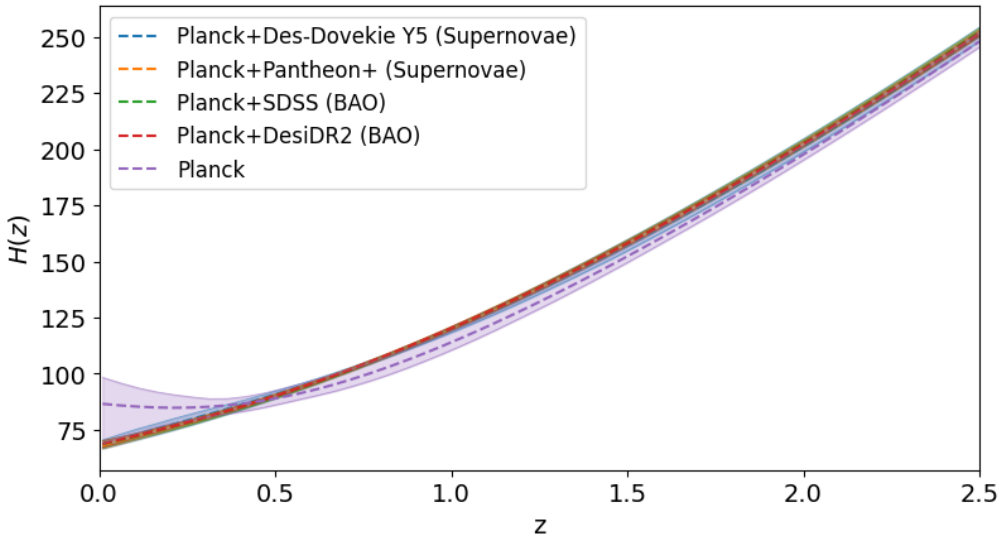


Figure 5: H(z) evolution in CPL with 2σ error bars.

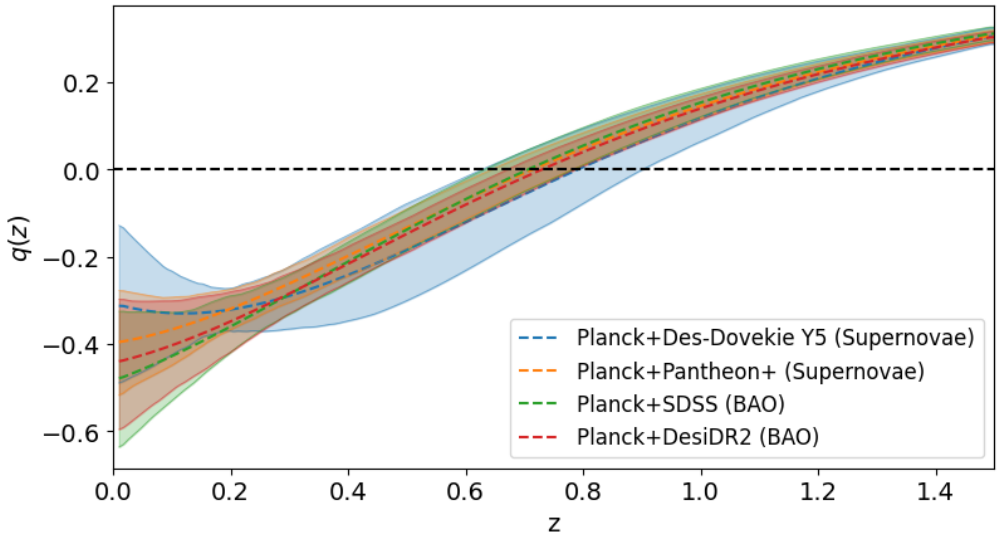


Figure 6: $q(z)$ in CPL with 2σ error bars

In this scenario, it is more convenient to add the Planck chain in Figure 5 in order to see how the evolution of the parameters is strongly influenced by the CMB likelihood. As a result, all the combinations that contain Planck exhibit almost the same behavior.

The dominant influence of the CMB likelihood on the $H(z)$ reconstruction is essentially given by two high-precision anchors. The acoustic peaks in the power spectrum of the anisotropies are measured with a precision of less than 1%, and the physics of which is determined by the baryonic and dark matter density parameters ($\Omega_b h^2$ and $\Omega_c h^2$). As the Hubble parameter evolution is primarily governed by Ω_m (See (37)), the power spectrum imposes a stringent constraint on the evolution of $H(z)$.

In addition, the angular size of the sound horizon θ_* fixes the comoving distance up to the last scattering redshift, defined as an integral that depends on $H(z)$. Then, in order to make the model compatible with CMB measurements, the Hubble parameter evolution must conform to a well-defined trajectory. Consequently, the CMB acts as a global geometric constraint, drastically reducing the allowed volume of the CPL parameter space and leading to a low-uncertainty evolution of $H(z)$.

The first aspect to analyze in Figure 6 is the behavior of DES Y5 SN at a low redshift. Its discrepancy with respect to the other ones comes from the absence of SNe Ia at $z < 0.1$. This is crucial to fix initial conditions, specially in the deceleration parameter, because in CPL its value at $z = 0$ is highly sensitive to w_0 and Ω_m, so little variations of these parameters cause significant deviations at low redshifts. DES Y5 SN has a high number of measurements at mid and high redshifts ($z > 0.5$), but the lack of an anchor that provides initial conditions leads not only to a possible discrepancy but also to wider error bars. Pantheon+ has SNe in the low redshift range, and in addition to its lower statistical uncertainty, it has smaller error bars across the entire range of study.

Other fact one can notice from DES Y5 SN is the 'hourglass' behavior around $z \sim$

0.2. This is essentially due to the high data density in this range, that compensates for the statistical uncertainty.

Pantheon+ also has higher error bars at low redshifts than in the middle and high range. While a low-redshift growth in the error bars is common to both supernova datasets, in this case the variation is substantially less pronounced than in the DES Y5 SN case. Pantheon+ has the possibility to reduce these error bars fixing the absolute distance through Cepheids. However, in this case, they are not included, and the study of their contribution is not among the objectives of the project.

In the range $z \sim 0.5 - 1.0$, the mean of DES Y5 SN is outside the other chains' 2σ regions. In principle, this is generally not significant under 3σ, and this may be caused by the limitations of this dataset in constraining $w(z)$, which results in a different evolution for $q(z)$.

BAO chains also exhibit a higher uncertainty at lower redshifts, but in this case it is caused for a different reason. As BAO likelihoods contain the three-dimensional position of galaxies, there is a minimum distance at which galaxies can be observed. In addition, as the volume of a local redshift shell is intrinsically small, there is not enough amount of sound-horizon pairs to constrain the model, increasing the uncertainty. As the redshift increases, there are more galaxies in each shell, so the error bars decrease. BAO datasets start becoming highly efficient from $z \sim 0.2$. Beyond this threshold, the sampled volume becomes large enough to effectively suppress shot noise and sample variance, allowing the acoustic scale to serve as a high-precision geometric probe.

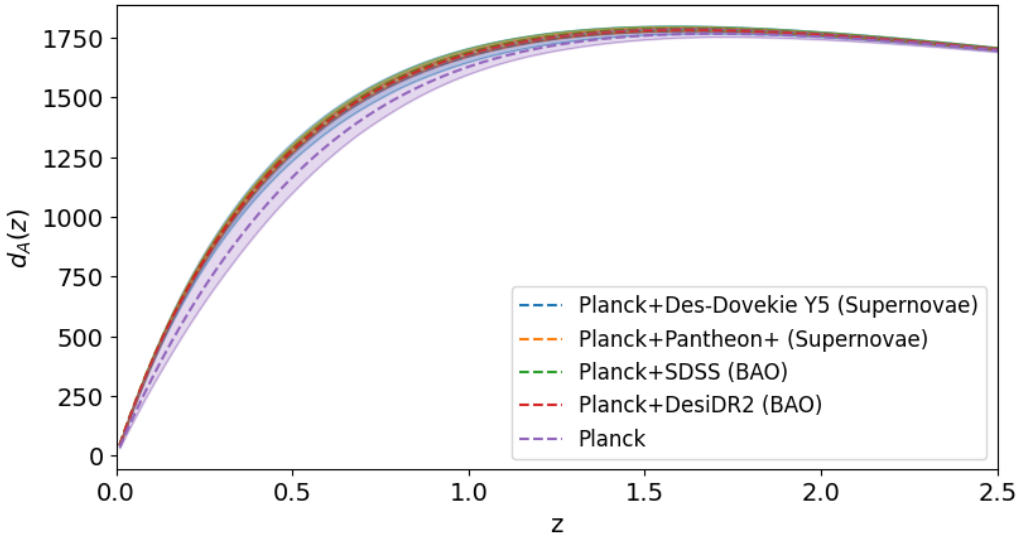


Figure 7: Angular diameter distance evolution in CPL with 2σ error bars.

The angular diameter distance evolution is also highly constrained by the CMB, as it is related to the comoving angular diameter distance by a factor $(1 + z)$, so determining $D_M(z_*)$ and $D_M(0) = 0$ limits how the parameter can evolve through the expansion history.

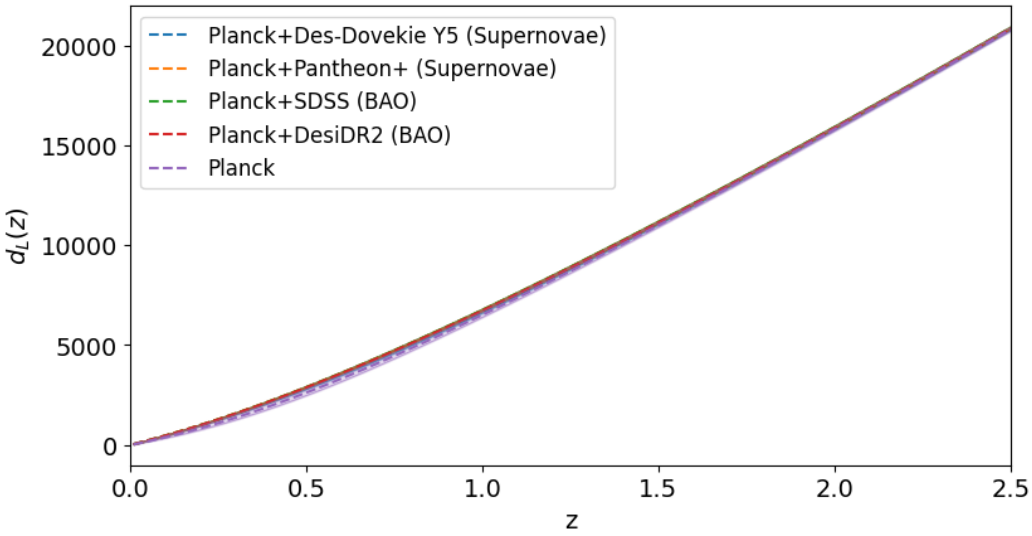


Figure 8: Luminosity distance evolution in CPL with 2σ error bars.

Mathematically, the comoving angular diameter distance is uniquely linked in a flat FLRW universe to the luminosity distance via the Etherington relation, $d_L(z) = (1+z)D_M(z)$. Consequently, any physical constraint placed into any of them is directly propagated to the other. That means the CMB likelihoods also represent the greatest constraint on the evolution of the luminosity distance.

However, the restrictions that the datasets impose on each of them vary depending on the redshift range. In low redshifts ($z \lesssim 0.15$), SNe Ia likelihoods provide a high density sample of $d_L(z)$, while BAO statistical volume is too low due to its spatial limitation to yield galaxy pairs. BAO surveys dominate at intermediate and high redshifts ($z \gtrsim 0.2$), not only due to its higher statistical volume but also because the BAO scale is absolutely calibrated by the CMB sound horizon r_s, giving also an absolute distance measurement of $D_M(z)$, whereas SNe Ia measurements only provide relative measurements if external calibrations are not used.

4.3 w_0-w_a plot for various datasets.

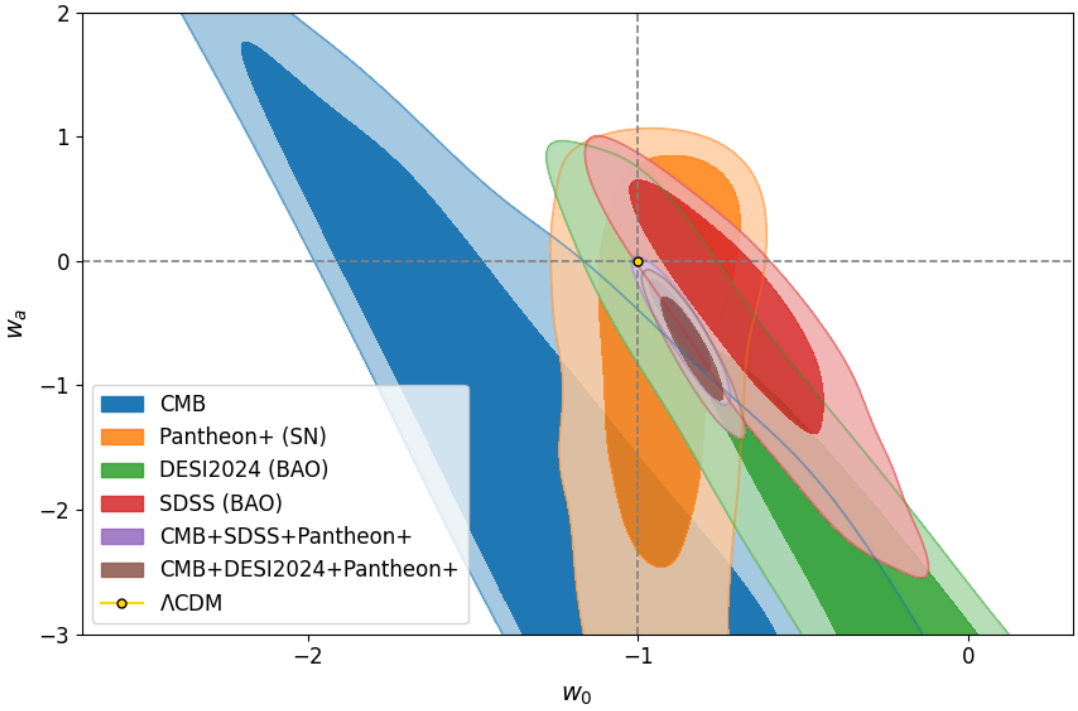


Figure 9: Covariance ellipses for $w_0 - w_a$ for different datasets. The crossed point $(w_0 = -1, w_a = 0)$ corresponds to ΛCDM.

To evaluate the constraining power of the different cosmological datasets under study, it is convenient to analyze the posterior distributions in the $w_0 - w_a$ plane. A notable feature in Figure 9 is the distinct orientation of the ellipses corresponding to Supernovae Ia compared to those of BAO and Planck. This difference in inclination is a direct consequence of the different redshift values at which their constraining power is maximum. This is called the pivot redshift, defined as the value at which the data constrains $w(z)$ with the minimum uncertainty, and it depends on the quality and the nature of each dataset. As a consequence, each dataset restricts the possible combinations for these parameters in a different direction of the parameter space. Far from being a problem, the different rotation of each ellipse breaks the possible degeneracies, shrinking the allowed contour area when performing joint likelihood analysis.

Since SNe Ia datasets are dominated by low-redshift observations ($z \lesssim 0.3$), their primary constraining power lies at smaller redshifts. By contrast, BAO samples extend to higher redshifts ($z \sim 0.5 - 2.5$), yielding to higher pivot redshifts and rotating the degeneracy axis. Just as in previous analyses, the principal contribution of Planck comes by fixing the scale of the sound horizon. That is also the reason why the CMB covariance ellipse occupies a larger portion of the parameter space and could be the reason why it has a $2\sigma - 3\sigma$ uncertainty relative to the ellipses of the combinations.

Another interesting aspect is the different uncertainties of the ellipses of the com-

binations relative to the ΛCDM cross-point. Whereas Planck+SDSS+Pantheon+ contains the cross-point inside its 2σ ellipse, Planck+DESI2024+Pantheon+ lies beyond 2σ with respect to ΛCDM. While SDSS BAO distance measurements across $z \sim 0.3 - 1.0$ were fully consistent with the constant DE equation of state, the greater precision of the DESI2024 data causes a mild deviation, and when anchored with Planck's sound horizon and constrained at low redshift with Pantheon+, the cross-point is dragged out of the ellipse. This deviation indicates a slight preference for a DDE, and therefore points the way forward for future research to either accept or reject the alternative model.

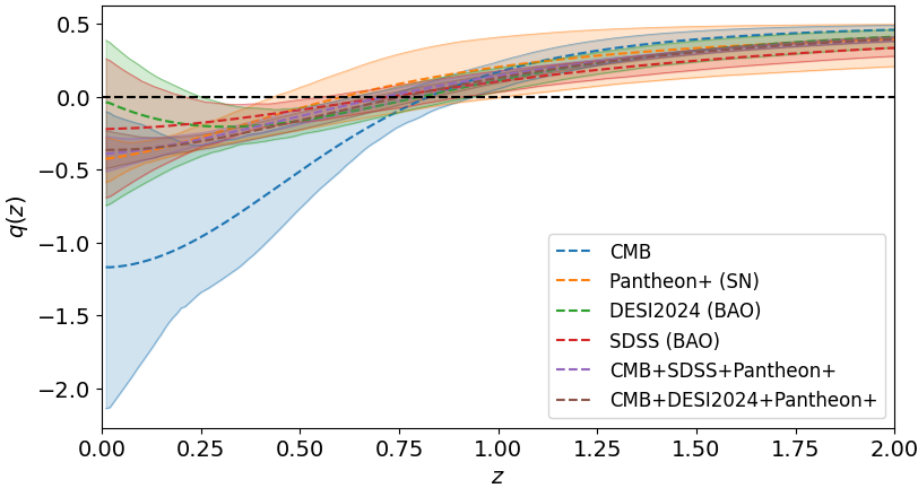


Figure 10: $q(z)$ evolution with 2σ error bars.

Unlike Figure 6, Figure 10 illustrates the redshift evolution of the deceleration parameter inferred from individual observables and their respective combinations, thereby enabling a clear evaluation of how each likelihood influences the overall constraints. The evolution of $q(z)$ performed by Planck in low redshifts is different from the other datasets, and its error bars are also the highest in this range. This behavior is due to the low constraining power of the CMB measurements on the late universe, just as in other examples studied. The mean of the deceleration parameter evolution of DESI2024 crosses back above 0 near $z = 0$, but this is not significant because the 2σ error bars in this range are similar to those from SDSS. In contrast, the late universe evolution from Pantheon+ has smaller uncertainties, and its evolution is similar to that given by the combinations, showing explicitly that the SNe Ia constitute the main constraining in this range.

In the mid range ($z \sim 0.5 - 1$), the BAO uncertainties shrink due to the higher data density. In contrast, SNe Ia error bars expand along the mid and high range, where the amount of data recovered is smaller. As $q(z >> 1) \to 1/2$, CMB error bars also shrink along the evolution toward the early universe. For the same reason, with the added fact that BAOs strongly determines the scale of distances, the combinations follow a very restricted evolution along the high redshift range.

5 Conclusions

The central objective of this work was to evaluate the statistical viability of dynamical dark energy (DDE) using the Chevallier-Polarski-Linder (CPL) parametrization, contrasting it with the standard ΛCDM null hypothesis ($w = -1, w_a = 0$). By performing Markov Chain Monte Carlo (MCMC) sampling with the Cobaya framework, we systematically analyzed how different observational likelihoods shape the posterior parameter space. The main findings of this research are summarized as follows:

1. **Preference for Dynamical Dark Energy:** The joint analysis of CMB, SNe Ia, and intermediate-redshift BAO data reveals a mild preference for a time-varying dark energy equation of state $w(z)$, deviating from the standard cosmological constant at up to the 3σ confidence level. The best-fit parameters systematically favor the region $w_0 > -1$ and $w_a < -1$, indicating a phantom-crossing behavior at redshifts around 0.4.

2. **Impact of Dataset Selection and BAO Likelihoods:** A comparative evaluation of BAO datasets demonstrates that dataset selection plays a crucial role and the results obtained by each one of them may lead to different conclusions. This illustrates how joint constraints on cosmological parameters remain highly susceptible to the specific calibration and errors of intermediate-redshift distance indicators.

3. **Parameter Degeneracies and Reconstruction:** The (w_0, w_a) confidence contours confirms that combining CMB data — which rigidly anchors the sound horizon scale r_s — with low- and intermediate-redshift probes significantly reduces the available parameter space, compressing the credibility intervals around the effective pivot redshift z_p.

The results obtained do not exclude ΛCDM, but they outline a compelling pattern that deserves further investigation. Upcoming data releases, such as DESI and future observations from Euclid and the Vera C. Rubin Observatory, will provide the statistical power required to determine whether the observed preference for dynamical dark energy corresponds to a true cosmological feature or a statistical fluctuation in current samples.

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Appendices

A Probability

Before entering in the core of this project, it is convenient to remember some notions of probability, since the algorithms used and the posterior analysis of the results are based on these principles [Trotta, 2017].

A.1 Notion of probability and random variables

When realizing an experiment, the total set of different results that can be obtained defines the sample space, denoted by Ω. Based on this, the classical (frequentist) definition of probability is the number of times an event occurs divided by the total number of possible events (those that make up the parameter space) in the limit of an infinite number of equiprobable trials.
The probability of two events A and B happening together is denoted as $P(A, B)$, and it is called the joint probability of A and B. The conditional probability of A given B is the probability of A given that B has already happened, denoted by $P(A|B)$. The sum rule for probabilities is

$$P(A) + P(\bar{A}) = 1, \tag{43}$$

where $\bar{A}$ denotes not A.

The product rule is

$$P(A, B) = P(A|B)P(B). \tag{44}$$

By inverting the order of A and B, the analogous relation can be obtained. From $P(A, B) = P(B, A)$, Bayes theorem can be obtained:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}. \tag{45}$$

A random variable (RV) is a function that transforms a random number from the parameter space to the space of real numbers. In other words, is the way to mathematise the events that occur in the real world.

A RV can be discrete, so there is only possible a countable number of outcomes (as in a coin tossing), or continuous, so there is a range of values of outcomes (as in a temperature measurement). Each RV has an associated probability distribution. For discrete ones, it is called probability mass function (pmf), giving a probability to each outcome: $P(X = x_i) = P_i = P(x_i)$ is the value of the RV X assuming it takes the value x_i. Its cumulative distribution function (cdf) is given by

$$C(x_i) := P(X \leq x_i) = \sum_{j=1}^{i} P(x_j). \tag{46}$$

For a continuous RV, the associated probability distribution is called probability

density function (pdf) $p(x)$, and its cdf is given by

$$P(x) := P(X \leq x) = \int_{-\infty}^{x} p(y)dy. \tag{47}$$

A.2 Central limit theorem

The Central Limit Theorem emphasises the importance of the Gaussian distribution in probability.

Theorem 1. *Let $X_1, X_2, ..., X_N$ be a collection of independent RV with a finite expectation value μ and a finite variance σ^2. Then, the sum for $N \to \infty$ is Gaussian distributed, and has mean $N\mu$ and variance $N\sigma^2$*

Theorem 2. *Let $X_1, X_2, ..., X_N$ be a collection of independent RV, each with finite expectation value μ_i and finite variance σ_i^2. Then, the variable*

$$Y = \frac{\sum_{i=1}^{N} X_i - \sum_{i=1}^{N} \mu_i}{\sqrt{\sum_{i=1}^{N} \sigma_i^2}} \tag{48}$$

is distributed as a Gaussian with expectation value 0 and variance 1.

A.3 The Likelihood function

Given a pdf or a pmf $P(X|\theta)$, where X represents a random variable and θ a collection of parameters describing the shape of the pdf and the observed data $\hat{x} = \{\hat{x}_1, \hat{x}_2, ..., \hat{x}_N\}$, its associated likelihood function $\mathscr{L}$ is defined as

$$\mathscr{L}(\theta) = p(X = \hat{x}|\theta), \tag{49}$$

meaning that the probability of observing the data $\hat{x}$ is a function of the parameters θ. Having defined the likelihood function, the Maximum Likelihood principle can be formulated: the set of parameters that maximizes the likelihood also maximizes the probability of obtaining the data. The Maximum Likelihood Estimator (MLE) is the set θ that maximizes the likelihood function:

$$\theta_{ML} \equiv \underset{\theta}{argmax}\, \mathscr{L}(\theta). \tag{50}$$

The MLE converges to the true value θ_{Real} for $N \to \infty$, and it is the estimator with the smallest errors. To find the MLE, the maximum value of the likelihood, i.e. where its first derivative is zero and its second derivative is negative, is studied:

$$\left.\frac{\partial \mathscr{L}(\theta)}{\partial \theta}\right|_{\theta_{ML}} = 0 \qquad \text{and} \qquad \left.\frac{\partial^2 \mathscr{L}(\theta)}{\partial \theta^2}\right|_{\theta_{ML}} < 0. \tag{51}$$

In practice, it is often used the logarithm of the likelihood, or log-likelihood, for example when programming, because it simplifies products as sums, and returns

values that are easier for the program to handle. As the logarithm is a monotonic function, maximizing the likelihood is the same as maximizing the log-likelihood.

$$\left.\frac{\partial ln\mathscr{L}(\theta)}{\partial\theta}\right|_{\theta_{ML}} = 0 \qquad and \qquad \left.\frac{\partial^2 ln\mathscr{L}(\theta)}{\partial\theta^2}\right|_{\theta_{ML}} < 0. \tag{52}$$

For a general likelihood function, a Taylor expansion around its maximum can be performed, given by θ_{ML}:

$$ln\mathscr{L}(\theta) = ln\mathscr{L}(\theta_{ML}) + \left.\frac{\partial ln\mathscr{L}(\theta)}{\partial\theta}\right|_{\theta_{ML}} (\theta - \theta_{ML}) + \frac{1}{2}\left.\frac{\partial^2 ln\mathscr{L}(\theta)}{\partial\theta^2}\right|_{\theta_{ML}} (\theta - \theta_{ML})^2 + ... \tag{53}$$

The second term vanishes by definition of the MLE, so the likelihood can be approximated as

$$ln\mathscr{L}(\theta) \simeq ln\mathscr{L}(\theta) exp\left(-\frac{1}{2}\frac{(\theta - \theta_{ML})^2}{\Sigma_\theta^2}\right) + ..., \tag{54}$$

where

$$\frac{1}{\Sigma_\theta^2} = -\left.\frac{\partial ln\mathscr{L}(\theta)}{\partial\theta}\right|_{\theta_{ML}}. \tag{55}$$

Then a general likelihood function can be approximated as a gaussian around its ML value, and the uncertainty around this value, Σ_θ, is approximately given by Eq. (55).

B Bayesian parameter inference

Bayes theorem becomes interesting when A is replaced by a set of parameters θ and B by the observed data or the samples, denoted as d. Then, Bayes Theorem can be written as

$$P(\theta|d) = \frac{P(d|\theta)P(\theta)}{P(d)}. \tag{56}$$

$P(\theta|d)$ is the posterior probability for θ, and it represents the degree of belief about the value of θ after seeing the observed data. $P(d|\theta) = \mathscr{L}(\theta)$ is the likelihood function, i.e. the probability of the data given a certain value for the parameters. $P(\theta)$ is called the prior, and it represents the degree of belief in the value of θ before seeing the data. When running an algorithm, the shape of the initial prior has to be chosen by the user, but the posterior generally converges to a prior-independent function for large datasets. The probability $P(d)$ is a normalizing constant, often called "evidence", defined as

$$P(d) = \int d\theta P(d|\theta)P(\theta). \tag{57}$$

Then, Bayes Theorem relates the information about the parameters after seeing the data to the likelihood and the prior, known before observing the data, updating the state of knowledge about a parameter as new data is introduced.

In the likelihood function, there are usually parameters that have an influence on the data but are not relevant. These parameters are called nuisance parameters ψ, and they can be eliminated by marginalization, i.e. by integrating over these parameters. As a consequence, the posterior pdf for the parameters of interest is given by

$$p(\theta|d) \propto \int \mathscr{L}(\theta, \psi) p(\theta, \psi) d\psi. \tag{58}$$

B.1 About the priors

In many situations, the choice of a correct prior is essential for achieving the right results. There are no general rules about choosing a correct one, but some aspects have to be considered. For example, if the parameter θ describes a quantity that must be in a certain range (e.g. the mass of a particle has to be positive), then the prior has to be 0 for values $\theta < 0$.
The most standard choice is to take a flat or uniform prior on θ, defined as

$$P(\theta) = \begin{cases} \frac{1}{\theta_{max} - \theta_{min}} & \text{for } \theta_{min} \leq \theta \leq \theta_{max} \\ 0 & \text{otherwise.} \end{cases} \tag{59}$$

With this choice, the posterior becomes functionally identical to the likelihood up to a proportionality constant. However, in parameter spaces of high dimensionality, the flat prior is not usually a good choice. Sampling with a uniform prior along each direction $x_i \in [0, 1]$ of a D-dimensional hypercube leads to the radius $r = \left(\sum_{i=1}^{D} x_i^2\right)^{1/2}$ of the samples. As a consequence,in a high dimensionality space, samples are found on a thin shell, and as the volume of the hypercube is much larger than the volume of the hypersphere, most of the space is not sampled.
In general, under a change of variable $\psi = \psi(\theta)$, the new prior transforms according to

$$P(\psi) = P(\theta) \left| det \left(\frac{\partial \theta}{\partial \psi} \right) \right|. \tag{60}$$

For a flat prior in θ, if the variable transformation is non-linear, the prior is not flat in ψ.

B.2 Markov Chain Monte Carlo methods

The purpose of Markov Chain Monte Carlo (MCMC) methods is to construct a sequence of points or samples in parameter space. This is called a chain, and its crucial property is that the density of samples (the frequency in which the samples fall within certain regions of the parameter space) is proportional to the posterior. This allows constructing a map of the posterior distribution.

A Markov Chain is a sequence of random variables $\{X^{(0)}, X^{(1)}, ..., X^{(N)}\}$ such that the probability of the $(t+1)$-th element in the chain only depends on the value of the t-th element. They can converge to a stationary state, i.e. the chain does

not change with t, and the successive elements of the chain are samples from the posterior.

The generation of new elements of the chain is described by the transition probability $T(\theta^{(t)}, \theta^{(t+1)})$, the probability of moving from $\theta^{(t)}$ to $\theta^{(t+1)}$ in parameter space. The Markov Chain also satisfies the detailed balance condition:

$$\frac{T(\theta^{(t)}, \theta^{(t+1)})}{T(\theta^{(t+1)}, \theta^{(t)})} = \frac{p(\theta^{(t+1)}|d)}{p(\theta^{(t)}|d)}, \tag{61}$$

i.e. the ratio of the posterior probabilities are inversely proportional to the ratio of the transition probabilities at the two points. Once having the samples from the posterior pdf, one can obtain expectation values for the parameters. The posterior mean is given by

$$E[\theta] = \int P(\theta|d)\theta d\theta \simeq \frac{1}{M}\sum_{t=0}^{M-1}\theta^{(t)}. \tag{62}$$

In general, the expectation value of any function of the parameters can be obtained as

$$E[f(\theta)] \simeq \frac{1}{M}\sum_{t=0}^{M-1} f(\theta^{(t)}). \tag{63}$$

One of the simplest and most used MCMC algorithms is Metropolis-Hastings [Metropolis et al., 1953, Hastings, 1970]. This mechanism works as follows: At step i, the chain is at a point $\theta^{(i)}$, and the algorithm proposes a candidate point $\theta^{(c)}$ from the proposal distribution $q(\theta^{(i)}, \theta^{(c)})$, that might be for example a Gaussian of fixed σ centered around $\theta^{(i)}$. This distribution satisfies the symmetry condition, $q(x, y) = q(y, x)$. Then, the posterior at the candidate point is evaluated, $p_c = p(\theta^{(c)}|d)$, and the candidate point is accepted with probability

$$\alpha = min\left(\frac{p_c q(\theta^{(c)}, \theta^{(i)})}{p_i q(\theta^{(i)}, \theta^{(c)})}, 1\right) = min\left(\frac{p_c}{p_i}, 1\right). \tag{64}$$

Then, if the candidate point performs better than the current point, it is accepted and added to the chain, and if $p_c < p_i$, then there is a probability of $\frac{p_c}{p_i}$ for the point to be accepted. This detail allows the algorithm to escape from a local maximum and explore all the parameter space. This algorithm also satisfies the detailed balance condition, with a transition probability given by $T(\theta^{(t)}, \theta^{(t+1)}) = q(\theta^{(t)}, \theta^{(t+1)})\alpha(\theta^{(t)}, \theta^{(t+1)})$. Assuming $\alpha(\theta^{(t)}, \theta^{(t+1)}) < 1$, then $\alpha(\theta^{(t+1)}, \theta^{(t)}) = 1$, and the ratio of the transition probabilities is

$$\frac{T(\theta^{(t)}, \theta^{(t+1)})}{T(\theta^{(t+1)}, \theta^{(t)})} = \frac{q(\theta^{(t)}, \theta^{(t+1)})p(\theta^{(t+1)}|d)}{q(\theta^{(t+1)}, \theta^{(t)})p(\theta^{(t)}|d)} = \frac{p(\theta^{(t+1)}|d)}{p(\theta^{(t)}|d)}. \tag{65}$$

In practice, some aspects have to be taken account to ensure the MCMC has converged to a fair representation of the posterior.

1. The Markov process takes a certain period before converging into the equilibrium distribution. This is called burn-in period, and points obtained before are usually rejected.

2. MCMC is a local algorithm that can be trapped around a local maxima fo the posterior density. Then, it is highly recommended to experiment with different algorithms, considering their strength and weakness.

3. The standard criterion to ensure the chain has explored correctly the parameter space is the Gelman-Rubin convergence criterion R [Gelman and Rubin, 1992]. This statistic compares the variance into each chain with the variance between chains. If all the chains converged to the same result, the variance between chains should be much smaller than the total variance. In this case R is close to 1. In this context, the value of $R-1$ is usually shown, because it can be more easily interpreted as an error margin. The standard value to ensure the convergence of the chain is $R-1=0.01$, which will also be used in all the chains ran in this project. When comparing different models, it is important to ensure the measurements converged to the same level $R-1$.

The principal advantage of MCMC methods is its great performance in high-dimensional problems: in traditional methods, the number of steps the program takes scale rapidly with the number of dimensions. MCMC does not try to explore all the parameter space; it tries to converge to high-probability regions.

The uncertainty of a parameter can be mathematically complex to calculate with traditional methods. In MCMC, the probability distribution is directly given by projecting the posterior probability on the axis of the parameter of interest by marginalization (See Eq. (58))

Many cosmological parameters are correlated, and MCMC captures very well these degeneracies, letting see how does a parameter depends on the other.

Previous information about the knowledge of the parameters can be easily included in the sampling by the use of priors.

In parameter spaces with complex topologies, MCMC has the capacity to skip from a local maximum to the global maximum thanks to the Metropolis Hastings algorithm, explained above.

B.3 Fast-Slow dragging

This is an algorithm designed specifically for Cobaya [Lewis, 2013]. Its objective is to explore faster the parameter space and to reduce the computational cost. First, parameters are divided into two blocks; fast and slow ones, depending on the time each parameter takes to take a step. Then, for each step taken by the slow block, keeping the slow block static, the algorithm takes many steps only for the fast block, allowing a faster convergence. The use of fast-slow dragging is especially important when the time that a slow parameter takes for each step is much higher than the time a fast parameter does. In cosmology, when some parameters are changed, the code has to solve a set of equations to ensure that the physical laws are followed, resulting in a much higher computational cost. In this scenario, as fast and slow parameters are correlated, allocating more computation weight to the fast parameters allow them to adapt to changes in slow ones, so that when the slow parameters take another step, they do so based on a set of parameters that has sampled efficiently.

C Cobaya

Cobaya [Torrado and Lewis, 2021] (Code for Bayesian Analysis) is a Bayesian sampling code written in Python and designed specifically for high-precision astrophysics and cosmology. It was principally developed by Jesús Torrado and Anthony Lewis (one of the creators of CosmoMC) with the aim of adapting to big-data cosmology. Mathematically it works as an ultra-efficient structure divided into three major components:

1. The sampler: the main objective of Cobaya is to explore the parameter space of the selected model. To do this, it uses an optimized MCMC algorithm with Fast-Slow Dragging, allowing a faster convergence.

2. Boltzmann solvers: Cobaya does not compute relativistic or perturbation equations directly. Instead, it passes each proposed parameter point to a Boltzmann code, that integrates the coupled Einstein–Boltzmann linear perturbation equations, ensuring the physical validity of the predictions. The primary Boltzmann solvers in Cobaya are CAMB [Lewis et al., 2000] and CLASS [Blas et al., 2011].

3. The Likelihoods: once the algorithm has the prediction, it has to be compared with experimental observations. Each Likelihood calculates the χ^2 comparing the theory with the datasets, then Cobaya sums all the selected ones, and finally decides through Metropolis-Hastings acceptance criterion if the next point of the chain moves to a zone with a better fit.

When the chain reaches the convergence criterion, evaluated through the Gelman-Rubin indicator, Cobaya generates a set of files containing the samples obtained by the Markov Chain. In these files, each line contains the weight, the value of the log-posterior, the primary and derived parameters and the likelihoods from each step taken in the process. Cobaya also creates a backing file, useful to resume or repeat the chain, a register with details about the process of the execution, a progress file and a checkpoint file.

One of the main advantages of this code is that the selection of the model, the likelihoods, the priors, the solver and the parameters can be easily changed from a text file in .yaml format, which would otherwise require changing thousands of lines from the code for each run.

Finally, the results can be analyzed and compared using the Python library GetDist, useful for loading the chains and plot the covariance ellipses and the evolution of parameters of interest.